

Week 6

HEALTHCARE ANALYTICS

Power BI Dashboard & Data Modeling Project

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1. Introduction

This is the Week 6 report. This week we built a data model in Power BI and used it to make a dashboard.

- The data has patient details, hospital admissions, and billing records.
- We split one big table into smaller, connected tables.
- This makes the model easier to use and easier to build charts from.

2. Project Objectives

- Split the data into patient, admission, and billing tables.
- Connect the tables with the right relationships.
- Build measures for patient count, billing totals, average stay, and readmission rate.
- Build one dashboard page with filters.
- Check that every measure works correctly.

3. Dataset Overview

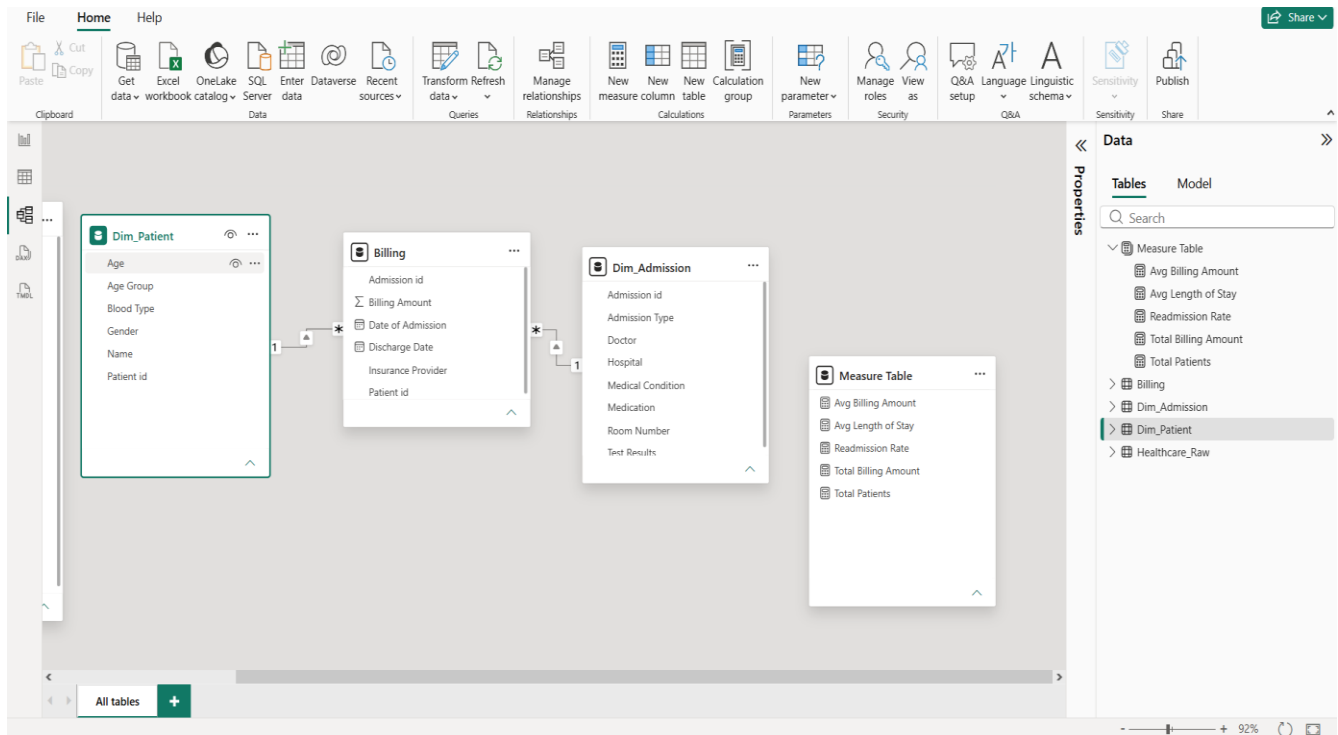
The raw data has one row per hospital visit. Each row has a patient, an admission, and a billing charge.

Table	Type	Key Fields
Dim_Patient	Dimension	Patient id, Name, Age, Age Group, Gender, Blood Type
Dim_Admission	Dimension	Admission id, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results
Billing	Fact	Admission id, Patient id, Billing Amount, Date of Admission, Discharge Date, Insurance Provider
Healthcare_Raw	Source	The original raw data, before cleaning
Measure Table	Measures	Holds all five DAX measures

We split this into three tables: Dim_Patient, Dim_Admission, and Billing. This avoids repeating the same details on every row.

4. Data Modeling & Schema Design

The model uses a simple star schema. Two tables connect to one central Billing table.



- Dim_Patient: Age, Age Group, Blood Type, Gender, Name, Patient id.
- Dim_Admission: Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results.
- Billing: Billing Amount, Date of Admission, Discharge Date, Insurance Provider, plus the two id fields that link everything.

5. Table Relationships

Each dimension table connects to Billing with a one-to-many relationship. This lets a filter (like age group) correctly filter the billing records.

From Table	Relationship	To Table	Cardinality
Dim_Patient	Patient id	Billing	1 : Many
Billing	Admission id	Dim_Admission	Many : 1

Both relationships filter one way only: from the dimension table into Billing. This keeps things simple.

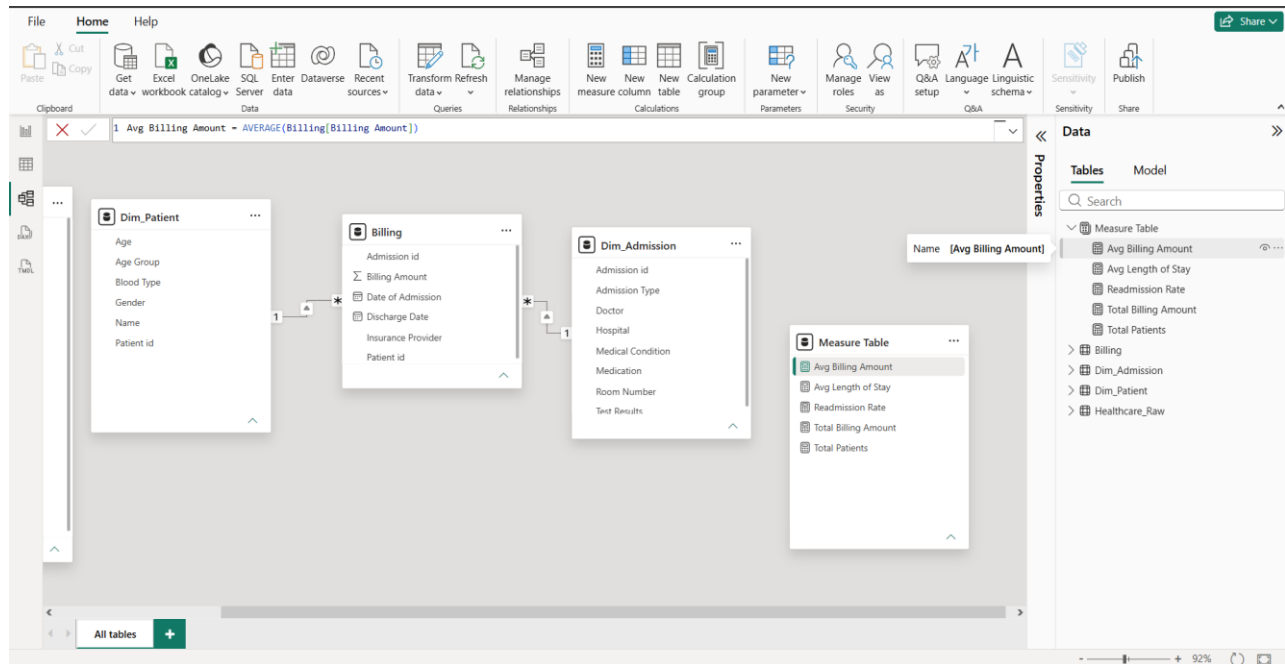
6. DAX Measures Developed

We built five measures on the Billing table and saved them in the Measure Table.

Measure	DAX Formula	Purpose
Total Patients	<code>COUNTROWS(Fact_Billing)</code>	Counts total patients.
Total Billing Amount	<code>SUM(Fact_Billing[Billing Amount])</code>	Adds up total revenue.
Avg Billing Amount	<code>AVERAGE(Fact_Billing[Billing Amount])</code>	Finds the average cost per patient.
Avg Length of Stay	<code>AVERAGEX(Fact_Billing, DATEDIFF(Fact_Billing[Date of Admission], Fact_Billing[Discharge Date], DAY))</code>	Finds the average number of days a patient stays.
Readmission Rate	<code>VAR TotalCount = [Total Patients] VAR RepeatPatients = COUNTROWS(FILTER(SUMMARIZE(Fact_Billing, Fact_Billing[PatientKey], "AdmissionCount", COUNTROWS(Fact_Billing)), [AdmissionCount] > 1)) RETURN DIVIDE(RepeatPatients, TotalCount, 0)</code>	Finds what share of patients were admitted more than once.

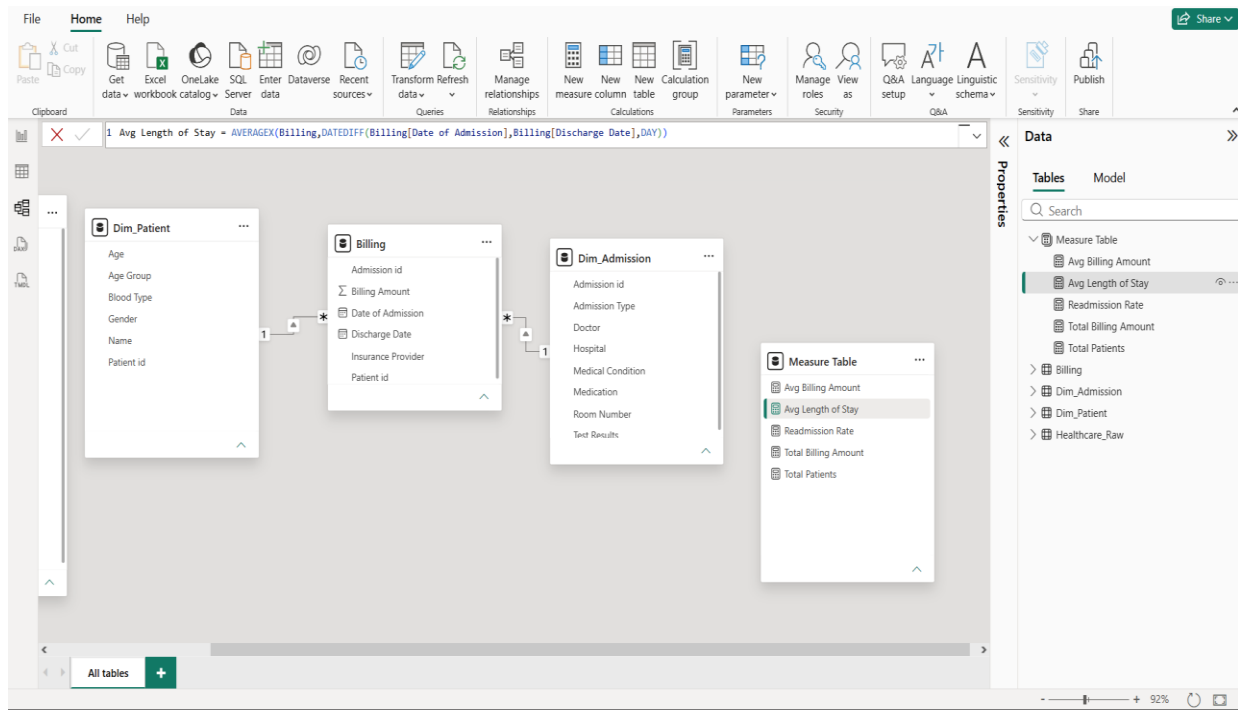
6.1 Avg Billing Amount

Average of the Billing Amount column. Shows the typical cost per admission.



6.2 Avg Length of Stay

Finds the days between Date of Admission and Discharge Date, then averages them. Shows how long a patient usually stays.



6.3 Readmission Rate

Counts total patients, then counts patients who show up more than once. Divides one by the other to get a rate.

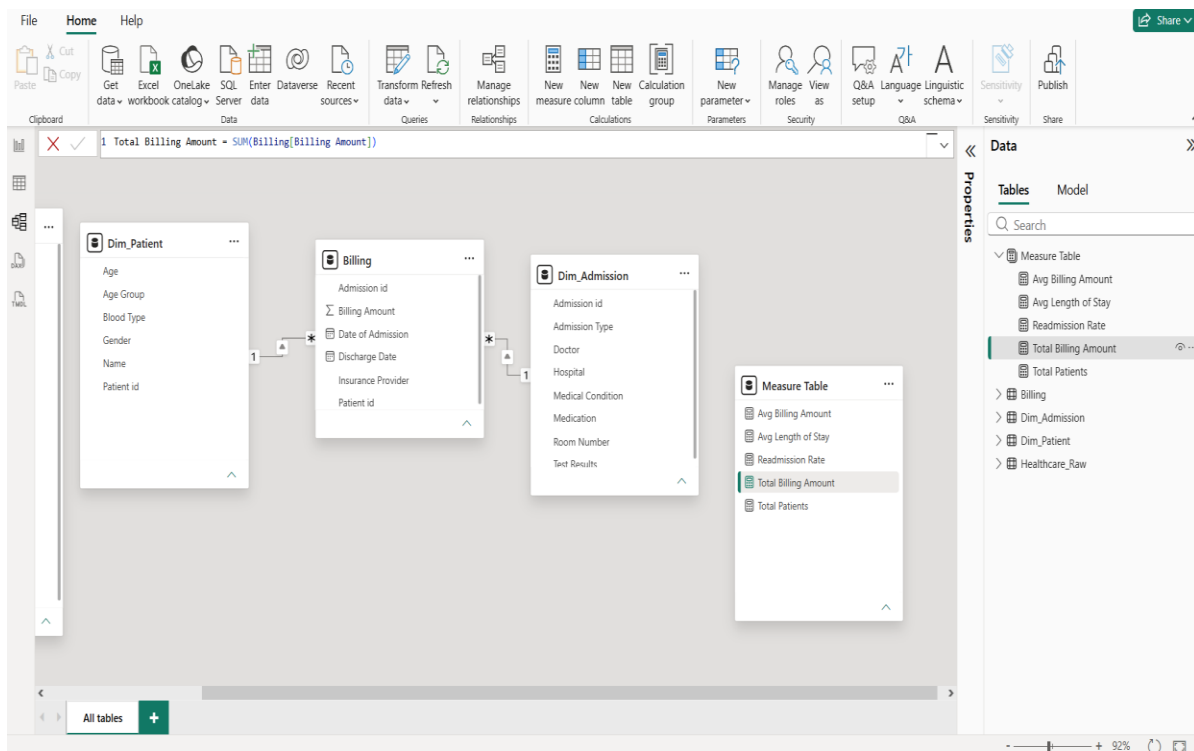
The screenshot displays the Microsoft Power BI Desktop interface. At the top, the ribbon shows the 'Home' tab with various toolbars including Clipboard, Data, Queries, Relationships, Calculations, Parameters, Security, Q&A, and Sensitivity. Below the ribbon, a DAX query is entered in the formula bar:

```
1 Readmission Rate = VAR TotalCount = [Total Patients]
2 VAR RepeatPatients = COUNTROWS(FILTER(SUMMARIZE('Billing','Billing'[Patient ID],"Admission Count",COUNTROWS('Billing')),[Admission Count] > 1))
3 RETURN DIVIDE(RepeatPatients,TotalCount,0)
```

The main workspace shows a data model diagram with three tables: Dim_Patient, Billing, and Dim_Admission. Dim_Patient is connected to Billing with a one-to-many relationship. Billing is connected to Dim_Admission with a one-to-many relationship. A 'Measure Table' is also shown, containing measures like Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, and Total Patients. The 'Readmission Rate' measure is highlighted. On the right, the 'Data' pane shows a search bar and a list of tables and measures, with 'Readmission Rate' selected.

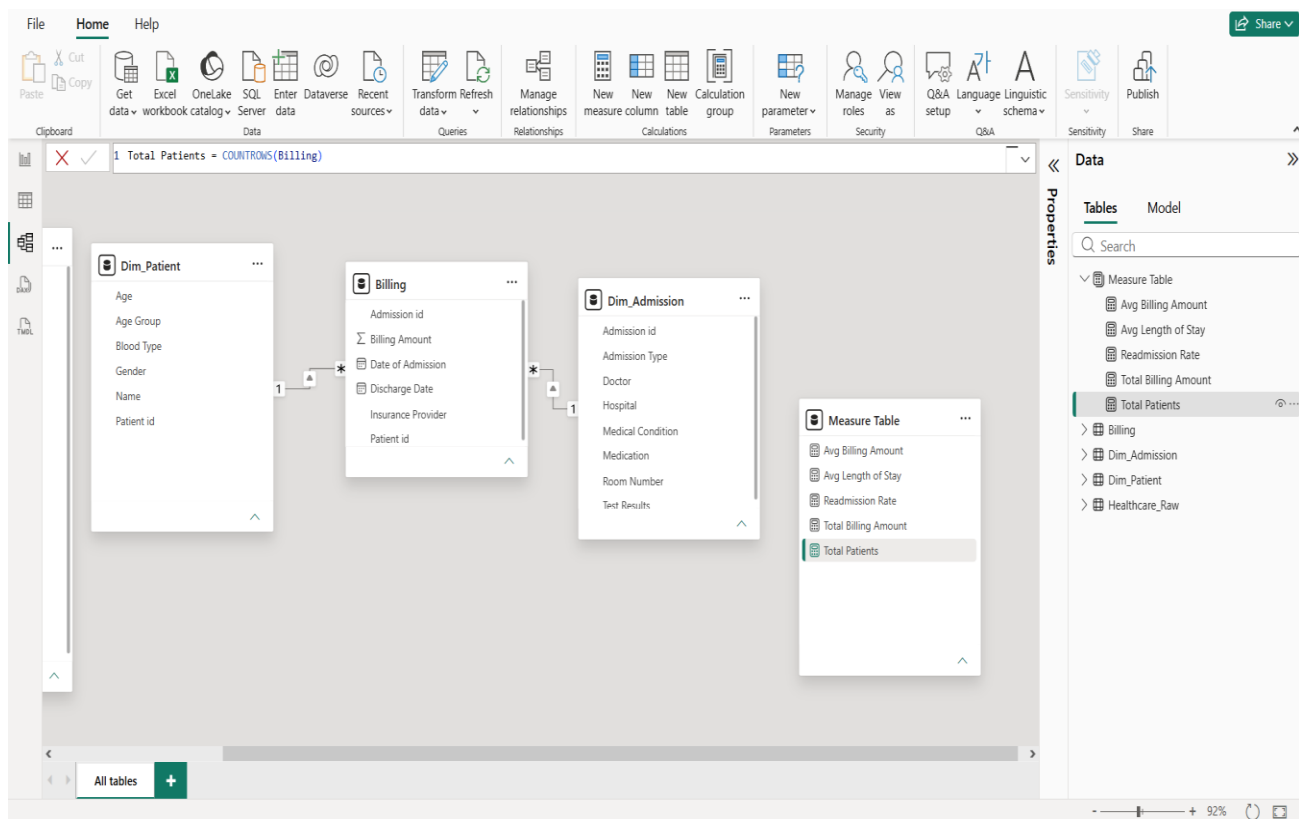
6.4 Total Billing Amount

Sum of the Billing Amount column. This is the total revenue.



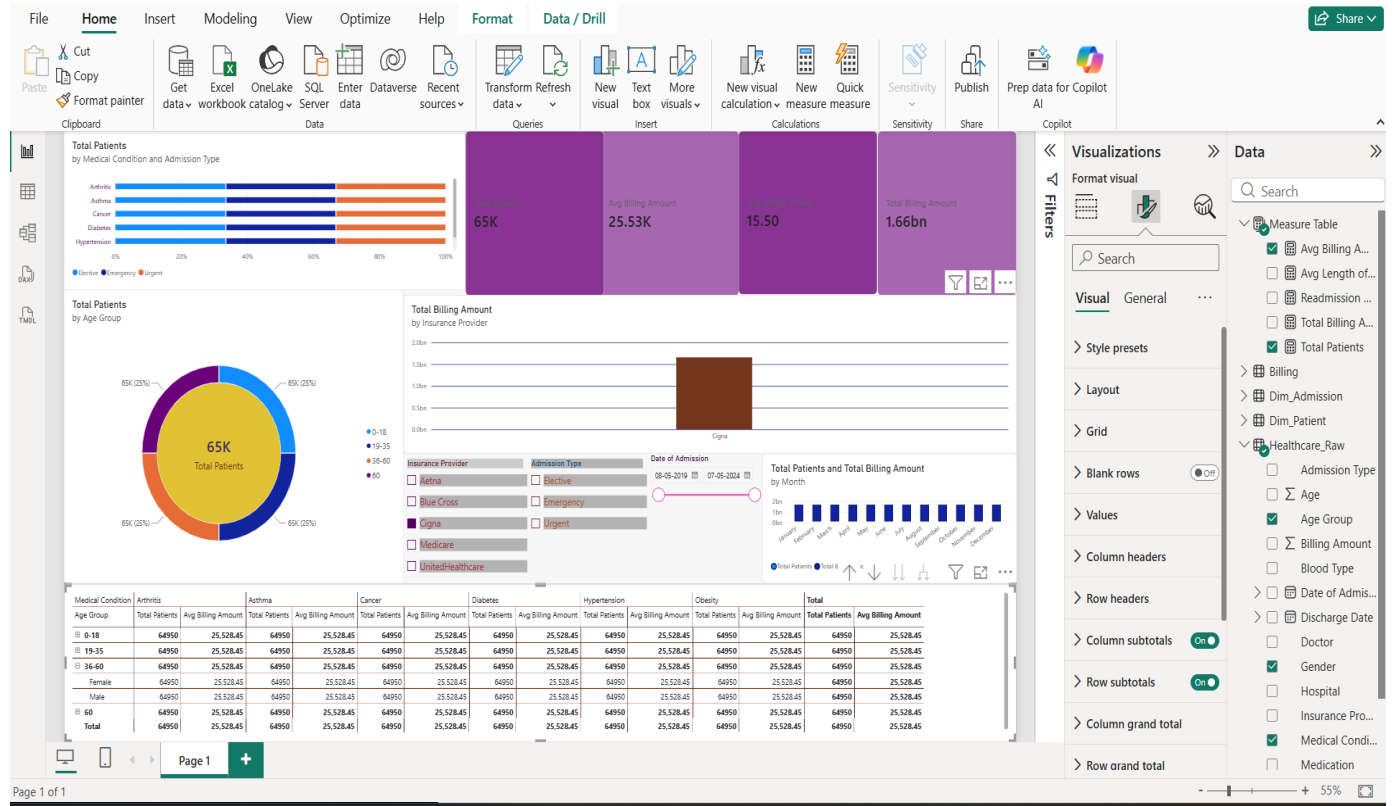
6.5 Total Patients

Counts the rows in the Billing table. Each row is one admission.



7. Dashboard Design & Visualizations

All five measures were placed on one dashboard page.



The dashboard has:

- KPI cards: Total Patients, Avg Billing Amount, Avg Length of Stay, Total Billing Amount.
- A bar chart: Total Patients by Medical Condition and Admission Type.
- A donut chart: Total Patients by Age Group.
- A column chart: Total Billing Amount by Insurance Provider.
- A trend chart: Total Patients and Total Billing Amount by month.
- A matrix table: Total Patients and Avg Billing Amount by Age Group and Medical Condition.
- Slicers: Insurance Provider, Admission Type, Date of Admission.

8. Key Insights

- Patients are spread evenly across all four age groups.
- Admission type looks similar across every medical condition.

- Billing amount and length of stay stay steady across groups.
- Cigna is the top insurance provider by total billing.
- Readmission Rate updates correctly with every filter.

9. Challenges & Learnings

- Readmission Rate was the hardest measure to build.
- DATEDIFF kept Avg Length of Stay as clean whole days.
- One Measure Table made all measures easy to find.
- One-way relationships kept the filtering simple.

10. Conclusion

This week we built a clean data model, five working measures, and one dashboard page. It gives a quick view of patient volume, cost, and readmissions.

- Next: add a date table for time-based measures.
- Next: add security so each hospital sees only its own data.
- Next: improve Readmission Rate to use a 30-day window.